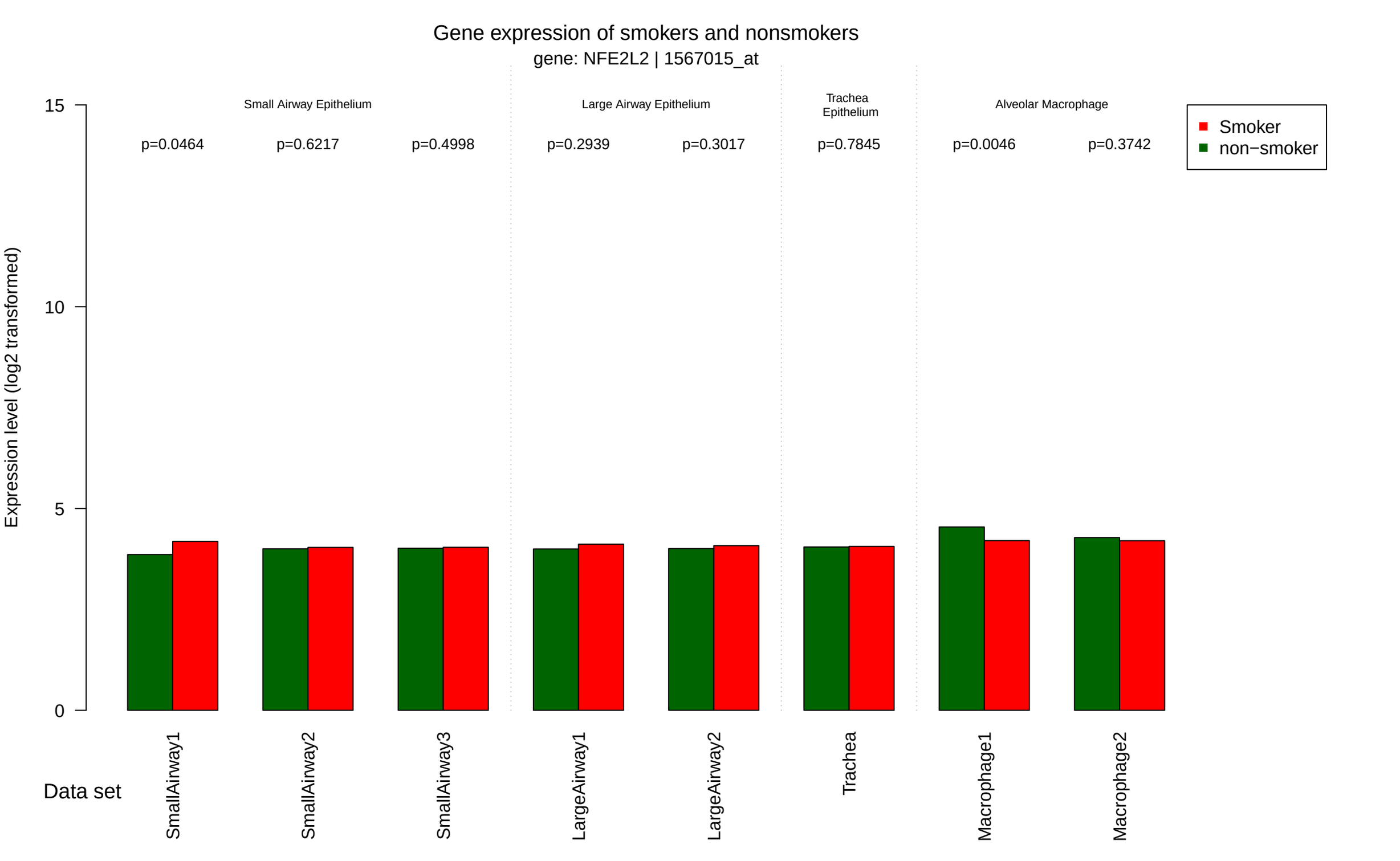
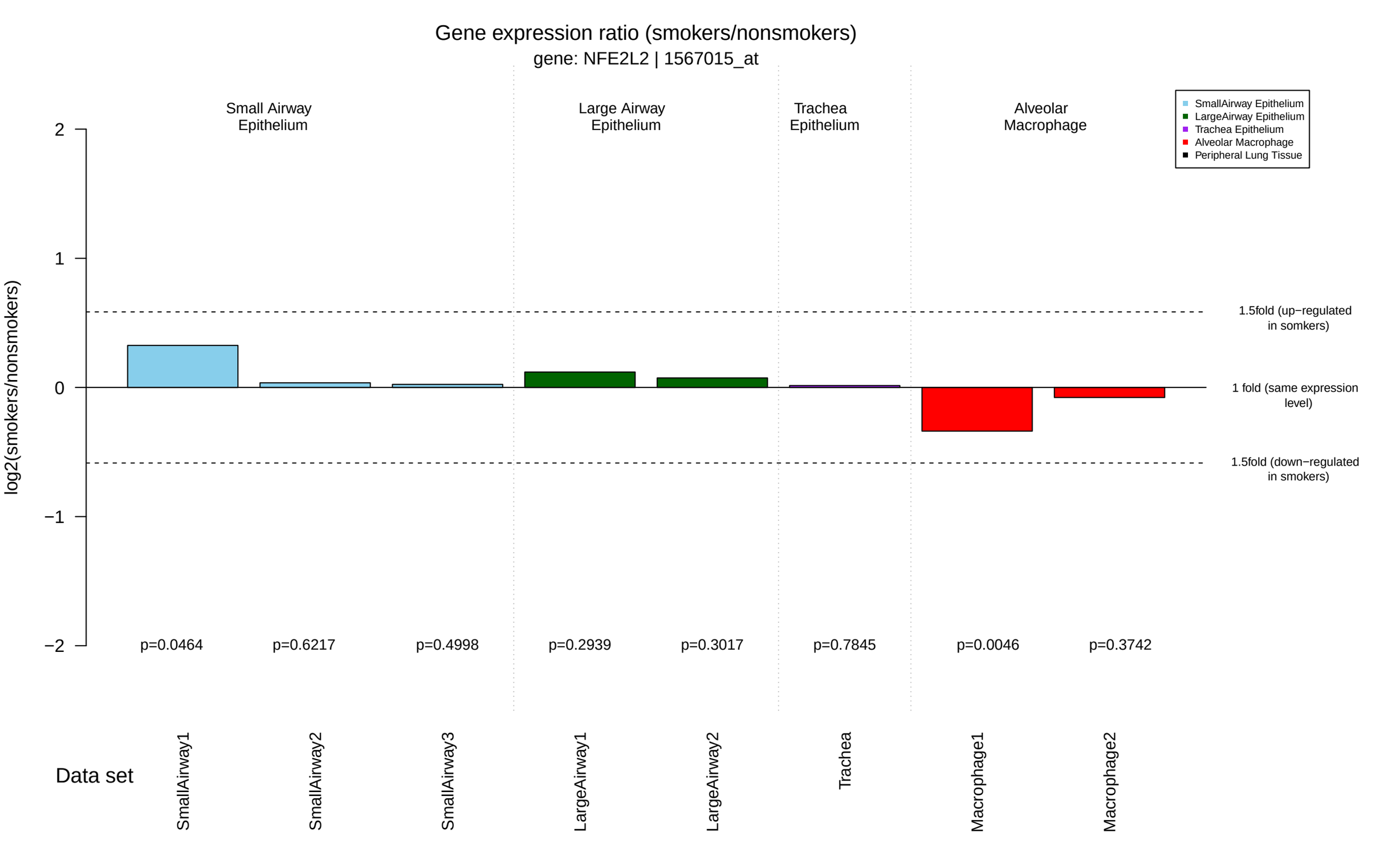
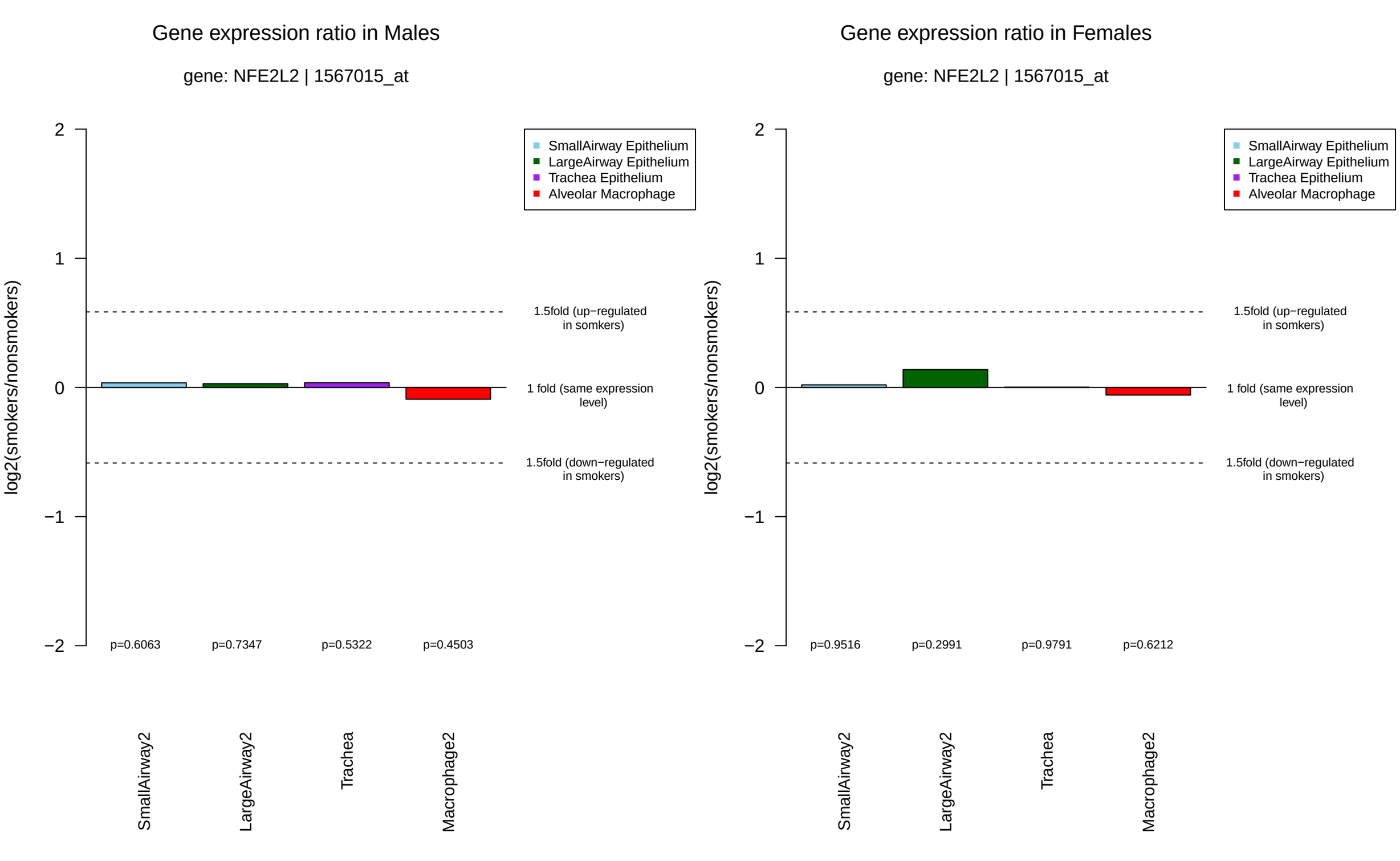
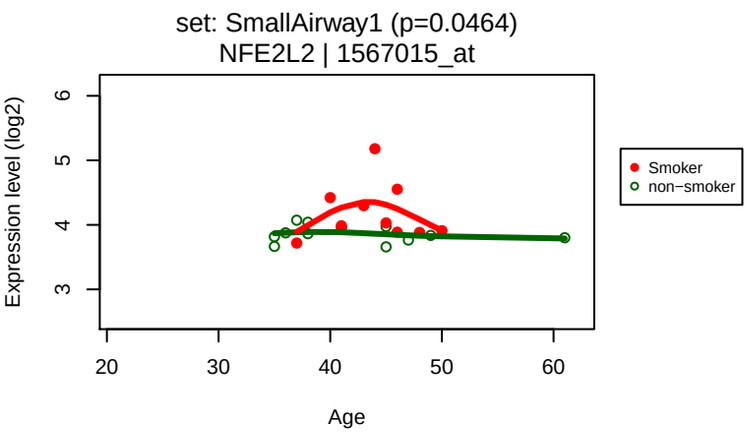
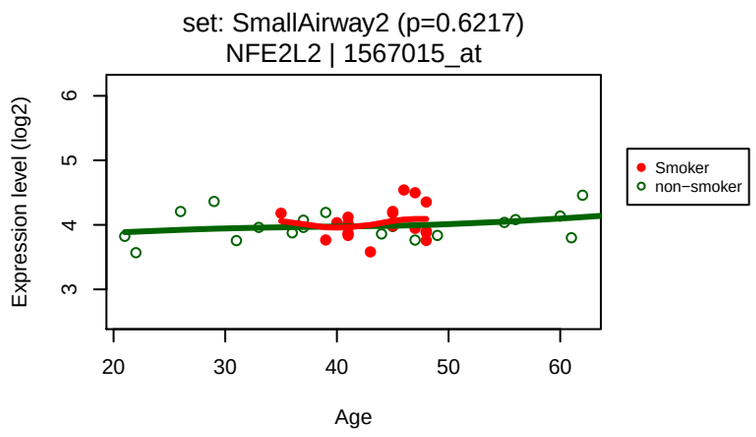
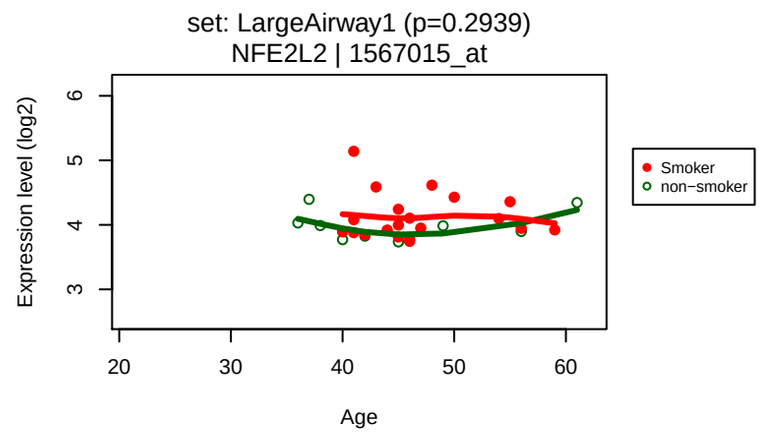

Gene expression of smokers and nonsmokers
gene: NFE2L2 | 1567015_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0464
p=0.6217
p=0.4998
p=0.2939
p=0.3017
p=0.7845
p=0.0046
p=0.3742
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

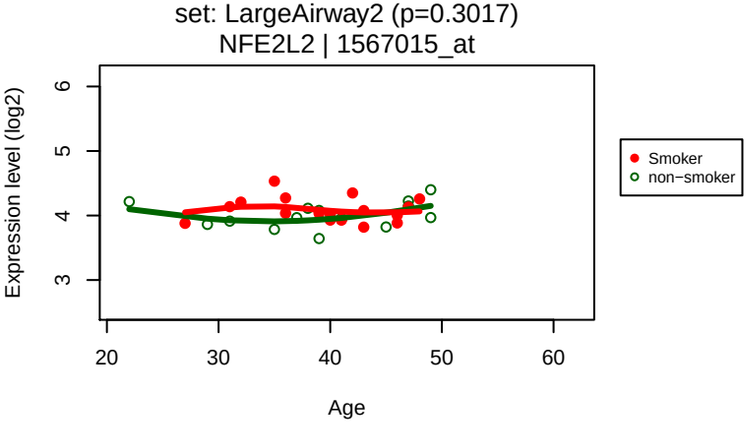

Gene expression ratio (smokers/nonsmokers)
gene: NFE2L2 | 1567015_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0464
p=0.6217
p=0.4998
p=0.2939
p=0.3017
p=0.7845
p=0.0046
p=0.3742
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

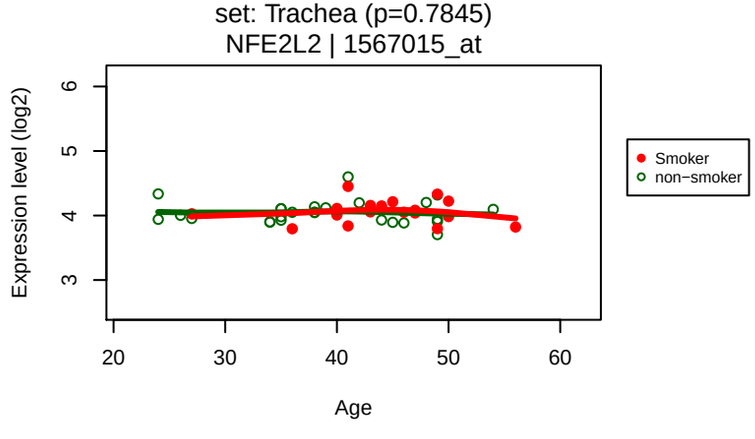

Gene expression ratio in Males
gene: NFE2L2 | 1567015_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6063
p=0.7347
p=0.5322
p=0.4503
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NFE2L2 | 1567015_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9516
p=0.2991
p=0.9791
p=0.6212
SmallAirway2
LargeAirway2
Trachea
Macrophage2

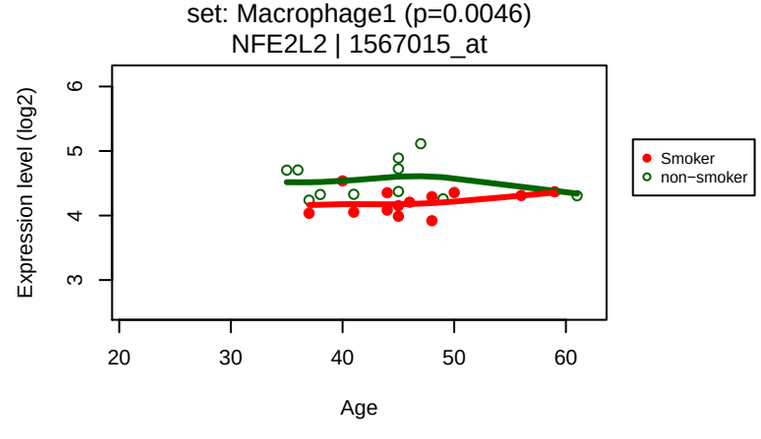

set: SmallAirway1 (p=0.0464)
NFE2L2 | 1567015_at
Expression level (log2)
Age
Smoker
non−smoker

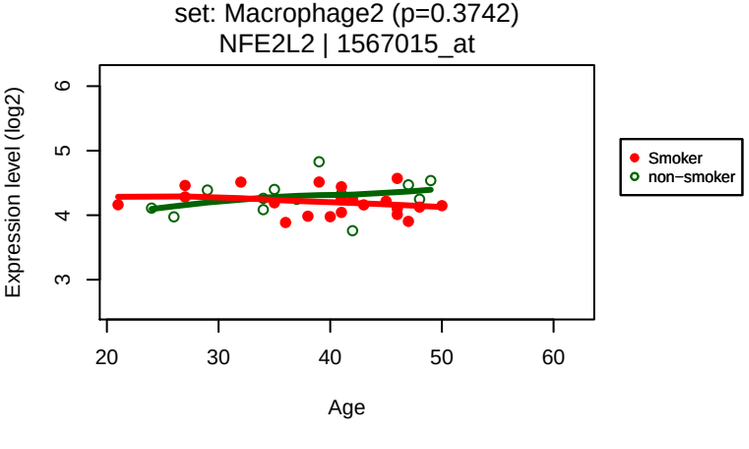

set: SmallAirway2 (p=0.6217)
NFE2L2 | 1567015_at
Expression level (log2)
Age
Smoker
non−smoker

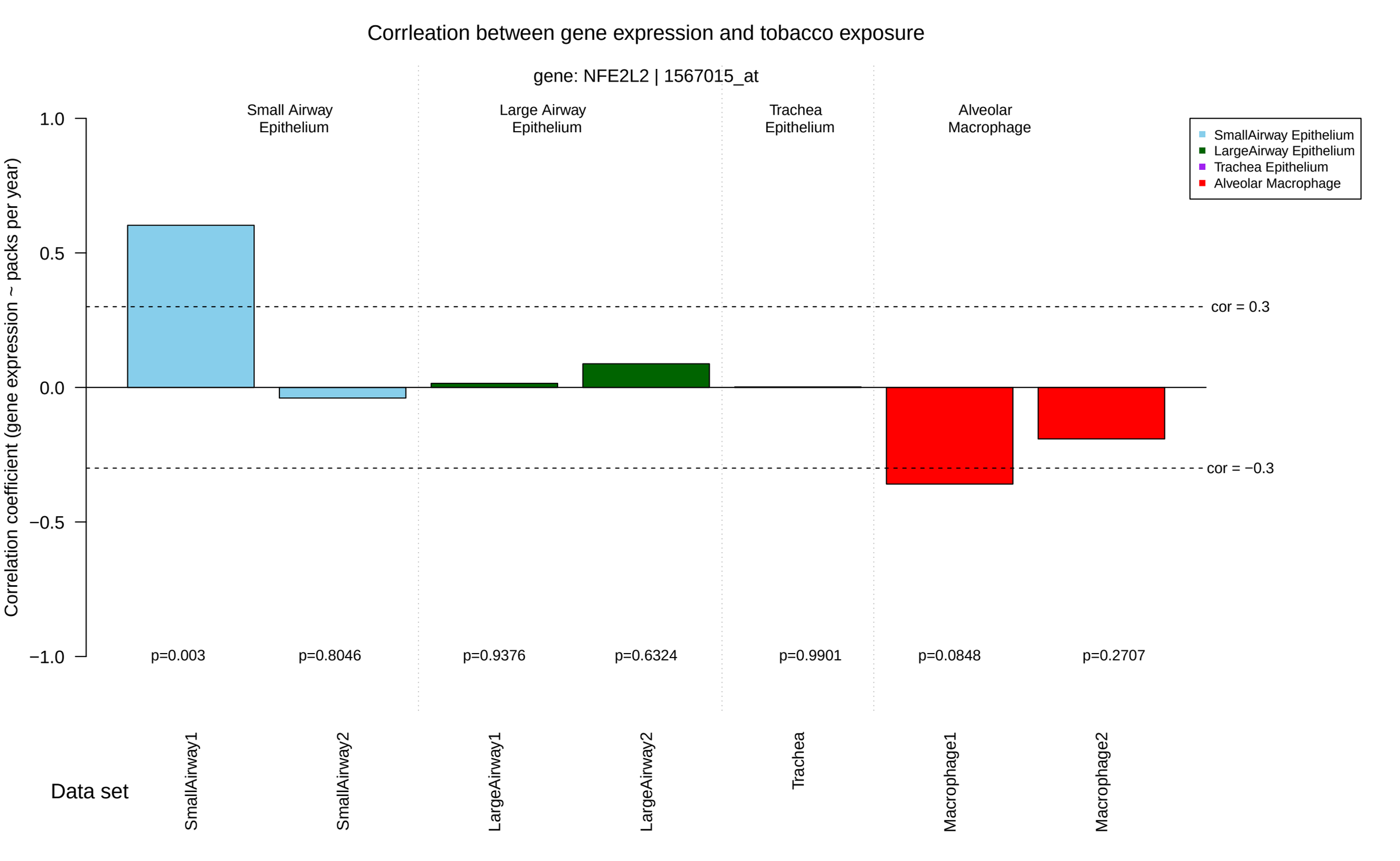

set: LargeAirway1 (p=0.2939)
NFE2L2 | 1567015_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3017)
NFE2L2 | 1567015_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7845)
NFE2L2 | 1567015_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0046)
NFE2L2 | 1567015_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3742)
NFE2L2 | 1567015_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NFE2L2 | 1567015_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.003
p=0.8046
p=0.9376
p=0.6324
p=0.9901
p=0.0848
p=0.2707
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2